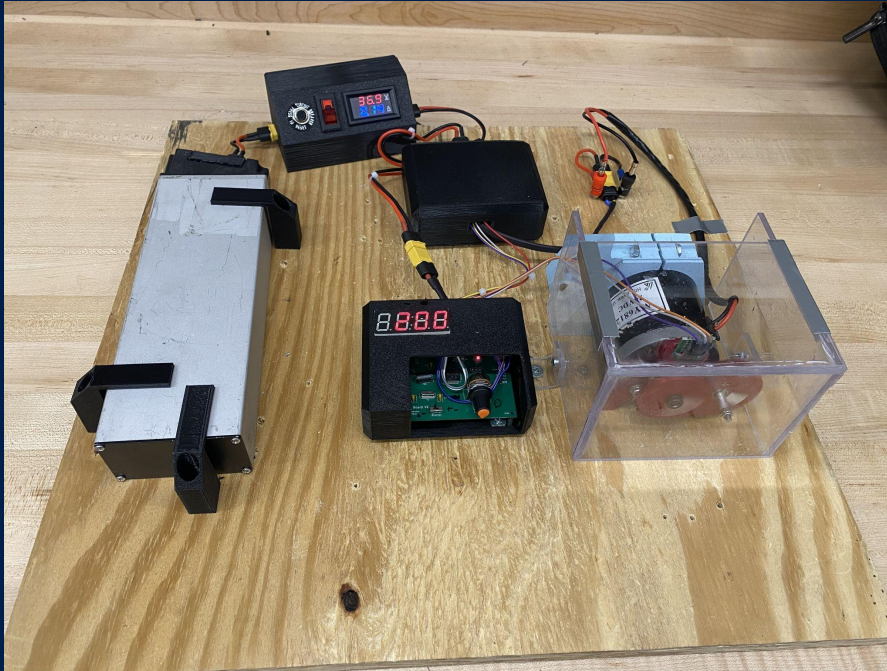


Figure 1



Electronic Speed Controller (ESC)

Amir Seidakhmetov & Joshua Wilbur



Demonstration

Presentation Forecast

1. Demonstration
2. Introduction
3. ESC Overview
4. Bootstrap Circuit Design
5. Conclusion
6. Questions

Purpose of an ESC

- ESCs are used to drive motors.
- Controls rotational speed and direction.
- Some ESCs incorporate feedback.
- Wide range of applications.



Figure 2 [2]



Figure 3 [1]

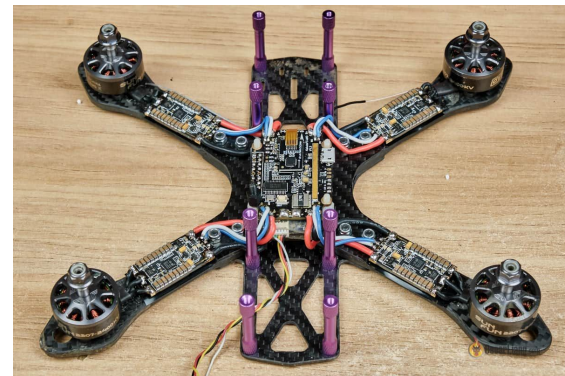


Figure 4 [3]

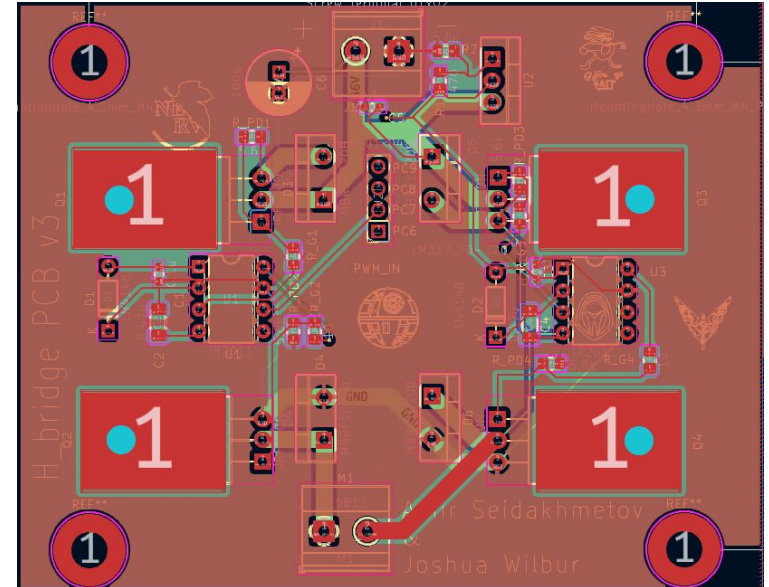
Why Did We Build an ESC?

- The team has an interest in electric longboards.
- Opportunity to design a complex control system.
- An ESC has real-world applications.

Figure 5 [3]



Figure 6



Specifications

- H-bridge circuit that can supply at least 2 amps to a motor.
- Closed-loop feedback system to monitor and adjust motor speed.
- Motor rotation speed ranging from 60 RPM $\pm$ 5% to 200 RPM $\pm$ 5%.
- Custom PCB designed for the project.
- All specifications were met during design review.

Figure 7 [11]



Similar Devices

- Hard to find exact comparisons.
- Piauda 36V/30A dual motor driver (\$99)
- Cytron MD25HV motor driver (\$59)



Figure 8 [5]



Figure 9 [6]



ESC Overview

Hardware Block Diagram

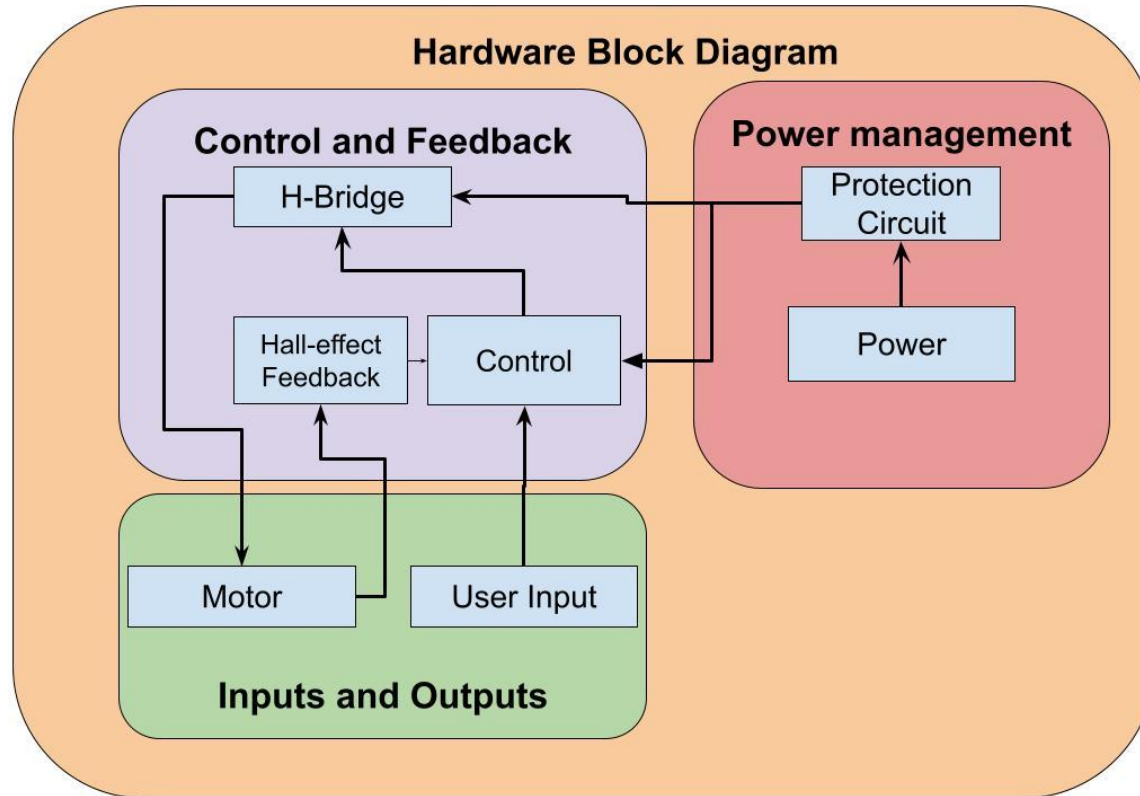


Figure 10

Software Block Diagram

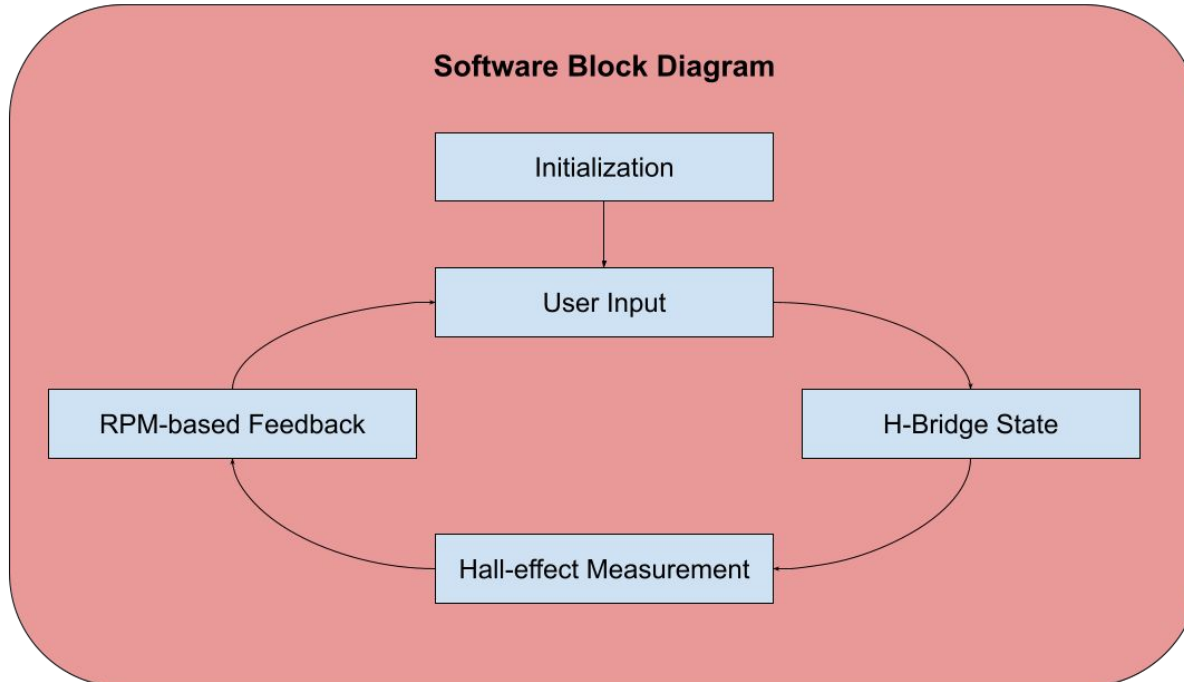


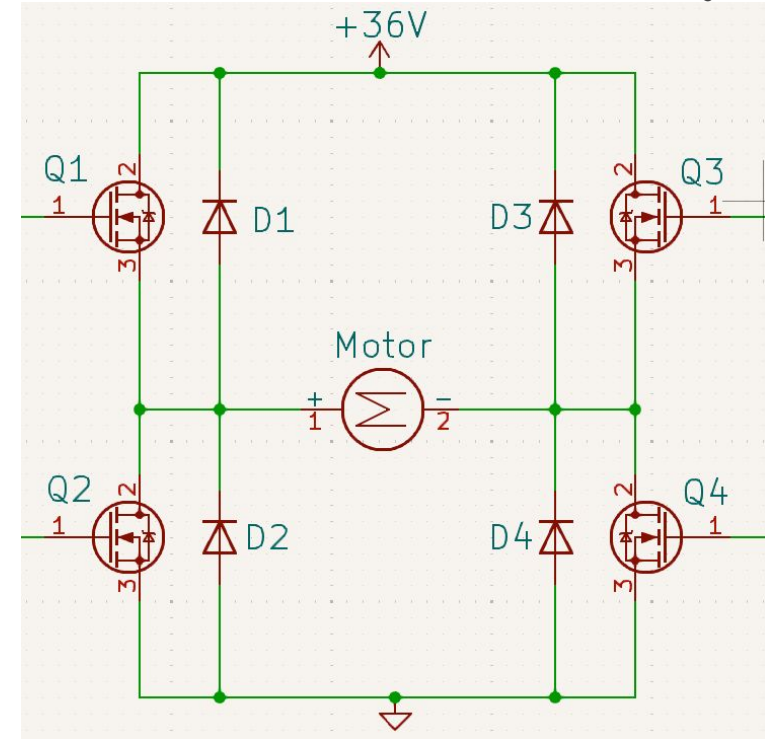
Figure 11



Bootstrap Circuit Design

Bootstrap Circuit Purpose

- Motor driven by a MOSFET H-bridge.
- MOSFETs require $V_{GS} > V_{th}$ to turn on
- $V_{DS} \geq (V_{GS} - V_{th})$ for saturation.
- Need a way to boost V_G above 36V supply.
- A bootstrap circuit can achieve this.



Gate Driver Selection

- Half-bridge driver for N-channel MOSFETs.
- Microcontroller outputs 3.3V signals.
- Gate current requirement.
- IR2011 from Infineon satisfied our needs.

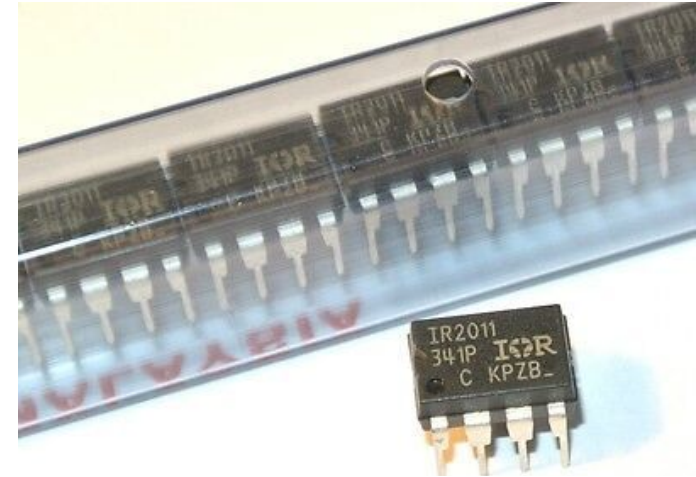
Figure 13

$$I_{G,max} = \frac{Q_G}{t_r}$$

$$I_{G,max} = \frac{37 \times 10^{-9}}{35 \times 10^{-9}}$$

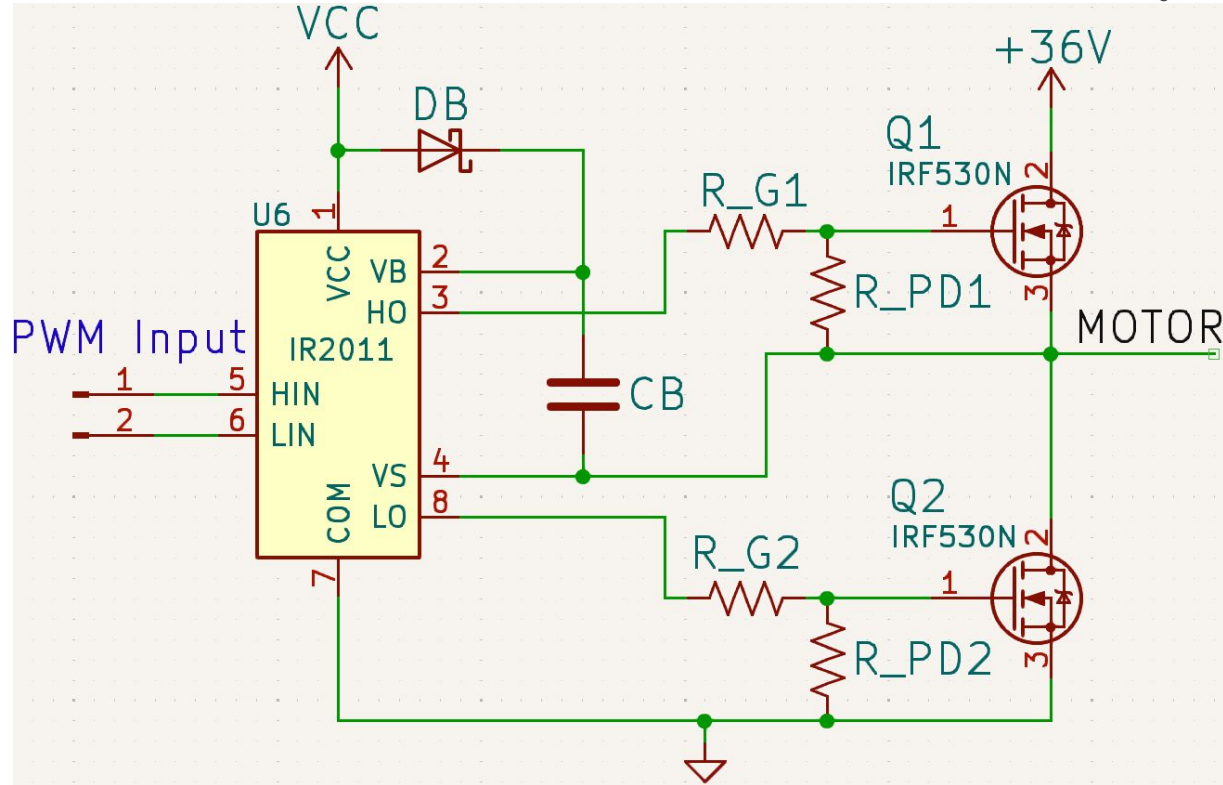
$$I_{G,max} = 1.05A$$

Figure 14 [7]



Bootstrap Circuit Components

Figure 15



Bootstrap Circuit Context

- Gate driver supply voltage (V_{CC}) set to 15V.
- $R_{DS(ON)}$ minimized when V_{GS} is 10V or greater.
- Bootstrap diode: 1N4148
- Voltage across C_B drops when high-side MOSFET turns on.



Figure 16

$$\Delta V_{BOOT} = V_{CC} - V_F - V_{GS}$$

$$\Delta V_{BOOT} = 15 - 0.7 - 10$$

$$\Delta V_{BOOT} = 4.3V$$

Bootstrap Capacitor Calculation

- Value dependant on total charge and allowable voltage drop.
- Recommended value is at least 10x larger than the minimum.
- The team used a 470nF capacitor.

Figure 18

$$Q_T = Q_G + Q_{LS}$$

$$Q_T = (37 + 3) \times 10^{-9}$$

$$Q_T = 40\text{nC}$$

Figure 19

$$C_G \geq \frac{10 \times Q_G}{\Delta V_{BOOT}}$$

$$C_G \geq \frac{400 \times 10^{-9}}{4.3\text{V}}$$

$$C_G \geq 93.02\text{nF}$$

Figure 20 [9]



Gate Resistor Calculations

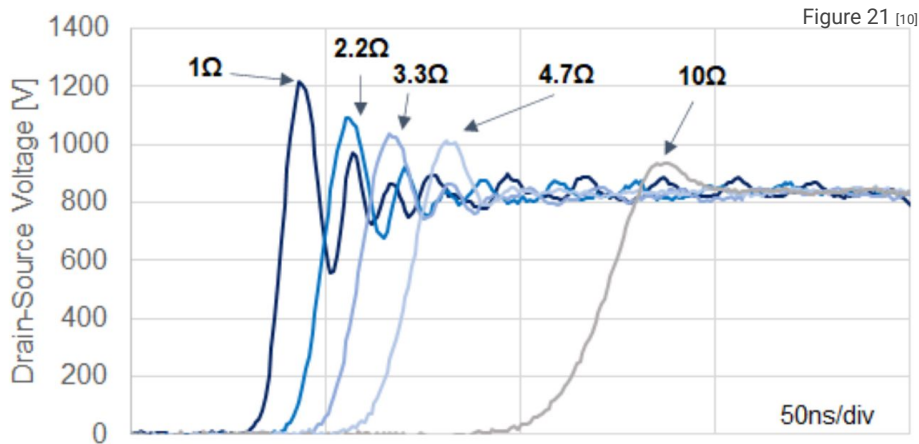
- Increased resistance slows switching transition.
- Mitigates ringing in the MOSFET gate.
- High-side gate voltage greater than low-side.

Figure 22

$$R_G = \frac{V_G}{I_{G,max}}$$

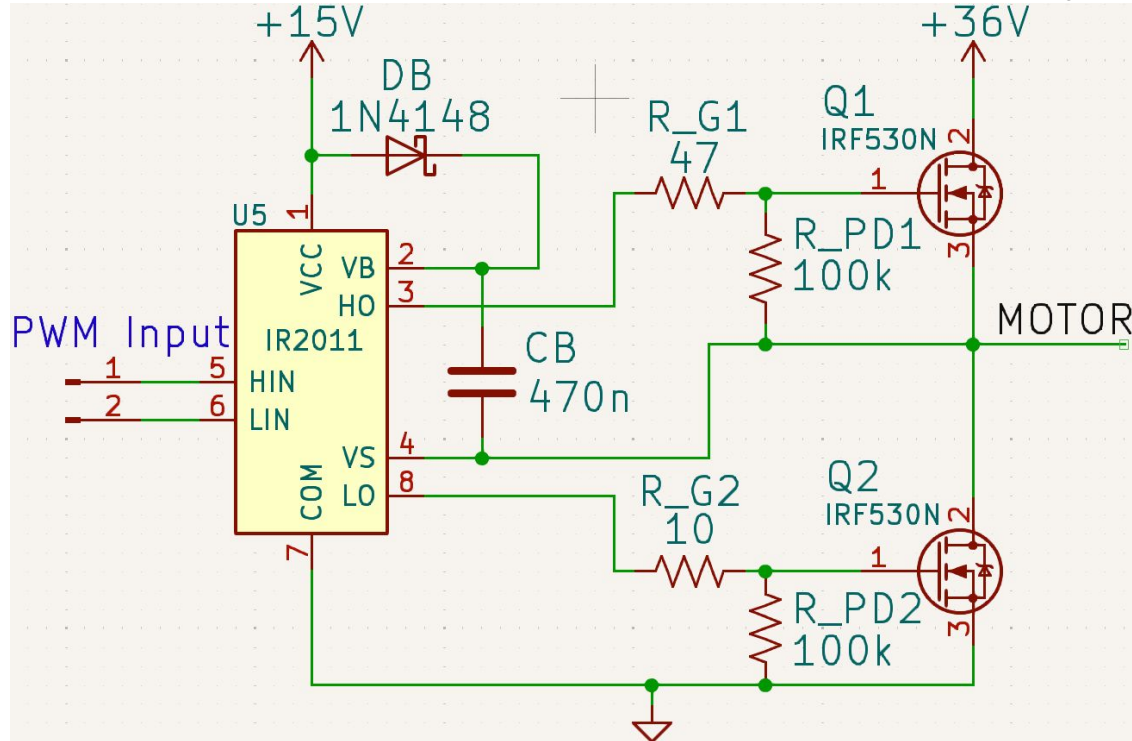
$$R_{G,high} = \frac{51}{1.05} = 48.57\Omega$$

$$R_{G,low} = \frac{15}{1.05} = 14.29\Omega$$



Designed Bootstrap Circuit

Figure 23





Conclusion

Specifications Testing

- Load test for H-bridge current specification.
- Tachometer used to verify speed.
- Manual loading to test feedback.

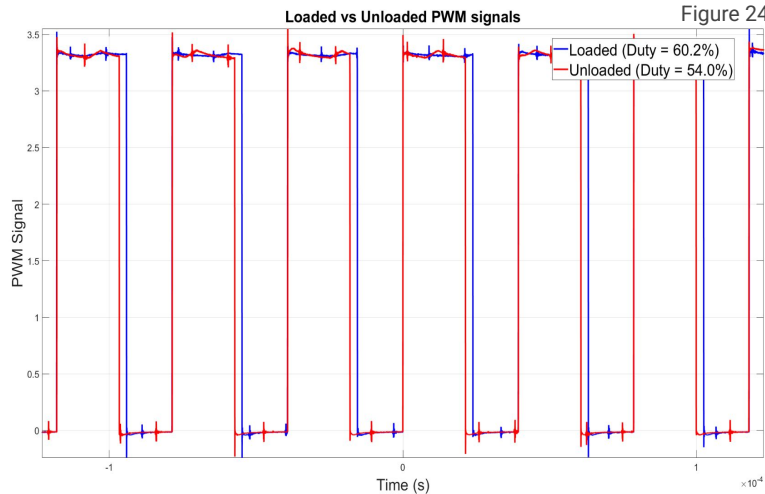
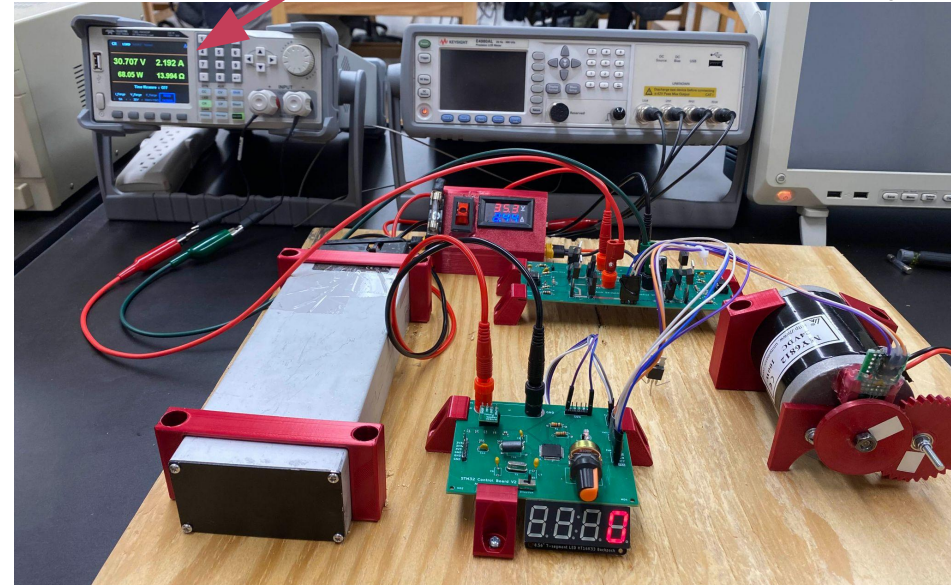


Figure 25



ESC Outcomes

- The designed ESC met the contract and all specifications.
- Average error in speed was 2.27%.
- Speed specification was limiting.
- The team is satisfied with the project outcome.

Figure 26

User Input	Positive Duty Cycle	Expected Speed	Measured Speed	Error
5%	45.27%	60 RPM	60.4 RPM	0.66%
15%	46.31%	75 RPM	78.7 RPM	4.93%
38%	48.59%	109 RPM	106.0 RPM	2.83%
55%	50.38%	135 RPM	132.5 RPM	1.85%
80%	53.20%	176 RPM	170.8 RPM	2.95%
100%	54.99%	200 RPM	200.9 RPM	0.45%

Advice to Juniors

- Carefully read through datasheets.
- Use available resources on campus.
- Learn PCB and CAD design.
- Don't doubt yourself too much, but do sanity checks
- Start early.
- Order extra parts.
- Seek help from professors.

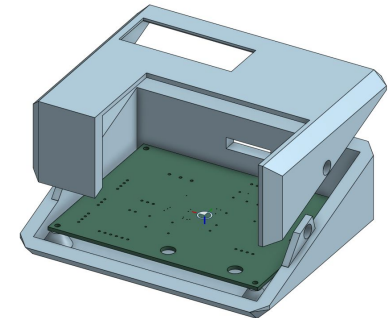
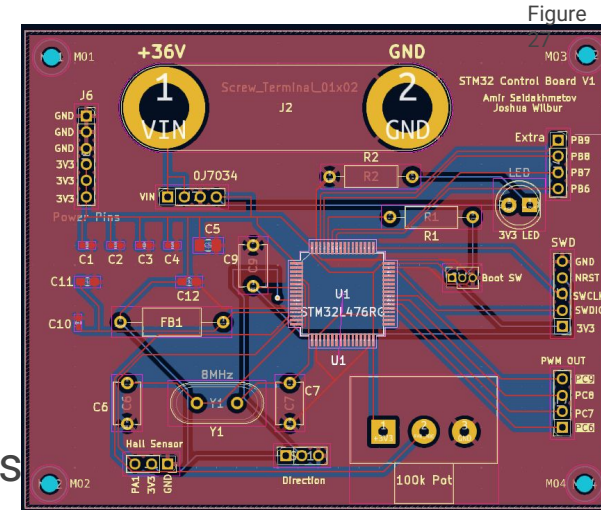


Figure 28



Questions

References

- [1] https://cdn.shopify.com/s/files/1/0071/3664/1135/products/3f78b11f-4b91-46de-8d47-90ac765680e1_2048x2048.jpg?v=1571709523
- [2] https://i5.walmartimages.com/asr/c25423d8-cd6b-4dfd-a03d-7d6a7bdeda9b_1.c19fde99052ac520eedd921e9d779c17.jpeg?odnWidth=1000&odnHeight=1000&odnBg=ffffff
- [3] <https://oscarliang.com/ctt/uploads/2017/12/how-to-build-fpv-mini-quad-racing-drone-solder-esc-to-fc.jpg>
- [4] https://supplyleader.com/preview_files/165.jpg
- [5] <https://www.robotshop.com/products/cytron-25a-7-58v-single-brushed-dc-motor-driver?qd=4d27e553cd49e3c18663a6d6c899faf2>
- [6] https://www.puaida.com/cdn/shop/files/20240509152312_1024x1024@2x.jpg?v=1715240759
- [7] <https://i.ebayimg.com/images/g/H4wAAOSwPhdU8hiP/s-l400.jpg>
- [8] <https://mm.digikey.com/Volume0/opasdata/d220001/medias/images/649/1N5229BFS.jpg>
- [9] <https://megatronica.cc/wp-content/uploads/2022/04/db25-22.jpg>
- [10] https://fscdn.rohm.com/en/products/databook/applinote/discrete/sic/mosfet/gate_drive_circuit-design_guidelines_an-e.pdf
- [11] <https://pixabay.com/photos/successful-consent-check-mark-hook-2633707/>